

NLP 201 HW 2

Ryan Okimoto

December 2025

1 Ngram

This section involves the implementation of an ngram model. This is done through by implementing a unigram, bigram, and trigram model.

1.1 Preprocessing

The first thing I did was create 3 different datasets for each ngram model. The unigram model would have no <START> token, a bigram would have a singular <START> token, and a trigram would have two <START> tokens and each would have a <STOP> appended at the end of each line in the dataset.

Any token that appears less than 3 times is turned into an <UNK> token

1.2 Perplexity

Formula:

$$PP = \exp \left(-\frac{1}{M} \sum \log P(\text{sentence}) \right)$$

M = total tokens INCLUDING <STOP> but NOT including <START>

For trigram model: First word after <START> uses bigram probability

1.3 Model Implementation

Unigram:

$$P(w) = \frac{\text{count}(w)}{\text{total_count}}$$

Bigram:

$$P(w_j | w_{j-1}) = \frac{\text{count}(w_{j-1}, w_j)}{\text{count}(w_{j-1})}$$

Trigram:

$$P(w_j | w_{j-2}, w_{j-1}) = \frac{\text{count}(w_{j-2}, w_{j-1}, w_j)}{\text{count}(w_{j-2}, w_{j-1})}$$

Special case: Trigram uses bigram probability for first word after <START>

1.4 Results and Expectations

We can expect our model to perform better with more context therefore trigram < bigram < unigram.

Model	Training	Development	Test
Unigram	976.54	892.25	896.50
Bigram	77.07	3443.04	3436.49
Trigram	7.87	2607451.93	2543381.81

Figure 1: Perplexity scores for unigram, bigram, and trigram models

2 Ngram with Smoothing

This section involves linear interpolation smoothing for ngram language models. This is done by combining unigram, bigram, and trigram models with weighted interpolation.

2.1 Linear Interpolation

$$P'(w_j | w_{j-2}, w_{j-1}) = \lambda_1 \cdot P(w_j) + \lambda_2 \cdot P(w_j | w_{j-1}) + \lambda_3 \cdot P(w_j | w_{j-2}, w_{j-1})$$

Where:

- $\lambda_1, \lambda_2, \lambda_3$ are weights that sum to 1
- $P(w_j)$ is the unigram probability
- $P(w_j | w_{j-1})$ is the bigram probability
- $P(w_j | w_{j-2}, w_{j-1})$ is the trigram probability

This was tested with the following lambda values:

- (0.3, 0.3, 0.4)
- (0.1, 0.3, 0.6)
- (0.2, 0.3, 0.5)
- (0.1, 0.4, 0.5)
- (0.15, 0.35, 0.5)

λ_1	λ_2	λ_3	Train PPL	Dev PPL
0.30	0.30	0.40	15.30	286.63
0.10	0.30	0.60	11.15	352.23
0.20	0.30	0.50	12.88	306.16
0.10	0.40	0.50	12.44	325.76
0.15	0.35	0.50	12.66	311.17

Figure 2: Lambda values and corresponding perplexity scores

2.2 Results

After running the model we found that when running the test set the best hyperparameters are $\lambda_1 = 0.3$, $\lambda_2 = 0.3$, $\lambda_3 = 0.4$ with perplexity = 286.63.

Then running with only half of the training data we get a perplexity value of 352.97 which is expected since there is less context for the model to train on.

And changing the unk threshold to 5 would likely decrease the perplexity since it is less likely to overfit to obscure words and better handle out of vocabulary words.

3 LLM Experimentation

This section experiments with OPT-1.3b for zero shot and few shot learning.

3.1 Experiment 1: Zero Shot Question Answering

We can see that while the model is pretty accurate it still makes mistakes and generates more complex answers than necessary.

3.2 Experiment 2: Few Shot Question Answering

We can see that the model is now performing much better as it has had a chance to learn on a couple examples before answering.

3.3 Experiment 3: Sentiment Analysis

We can still see that few shot prompting is performing better as more context will generally always correlate to better accuracy.

```

=====
Question: Who succeeded Joseph Willard as president?
=====
Generated Answer: Samuel Webber

=====
Question: When did David Tappan die?
=====
Generated Answer: Tappan died in 1803

=====
Question: Who was elected to the chair in 1805?
=====
Generated Answer: Henry Ware

=====
Question: What year did Joseph Willard die?
=====
Generated Answer: 1804
=====

```

Figure 3: Zero shot question answering results

```

1. Who succeeded Joseph Willard as president?
   Model Answer: Samuel Webber
   Expected: Samuel Webber

2. When did David Tappan die?
   Model Answer: Tappan died in 1803
   Expected: 1803

3. Who was elected to the chair in 1805?
   Model Answer: Henry Ware
   Expected: Henry Ware

4. What year did Joseph Willard die?
   Model Answer: 1804
   Expected: 1804

```

Figure 4: Few shot question answering results

```
-----  
Sentence: This movie was absolutely fantastic! I loved every minute of it.  
Predicted Sentiment: Positive  
  
-----  
Sentence: The food was terrible and the service was even worse.  
Predicted Sentiment: Negative  
  
-----  
Sentence: It was okay, nothing special but not bad either.  
Predicted Sentiment: Neutral  
  
-----  
Sentence: I'm so happy with my new phone, it works perfectly!  
Predicted Sentiment: Neutral  
  
-----  
Sentence: What a waste of money, I'm very disappointed.  
Predicted Sentiment: Neutral
```

Figure 5: Sentiment analysis results comparing zero shot and few shot